

Cryptography Algorithms and Verilog Implementation 130330

Verilog State Machines (Assignment 3)

Date: Sunday, March 22, 2026

Submission:	By not more than 2 people together
Deadline:	Sunday, April 19, 2026 (17:15) – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

The purpose of this lab is to exercise you with simulation and synthesis of Verilog state machines.

A. Verilog Design

Change the following state machine to give coffee only at a 5th coin and write Verilog code accordingly.

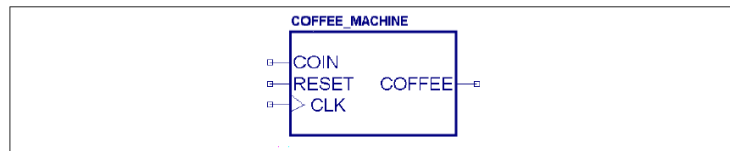
Example: MOORE Coffee-machine

- Description

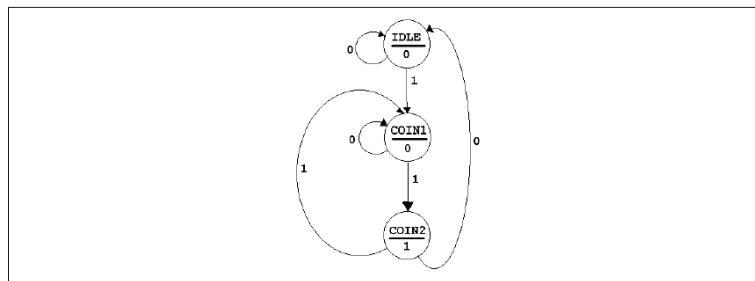
Inputs	COIN
Outputs	COFFEE
Characteristics	Two coins give coffee

- Synchronous machine (using a CLK signal) with synchronous RESET
- COIN and COFFEE are clock-cycle wide electrical logic pulses in an electro-mechanical coffee-machine.

- Symbol



- State diagram (MOORE coffee-machine)



B. Verilog Test Bench

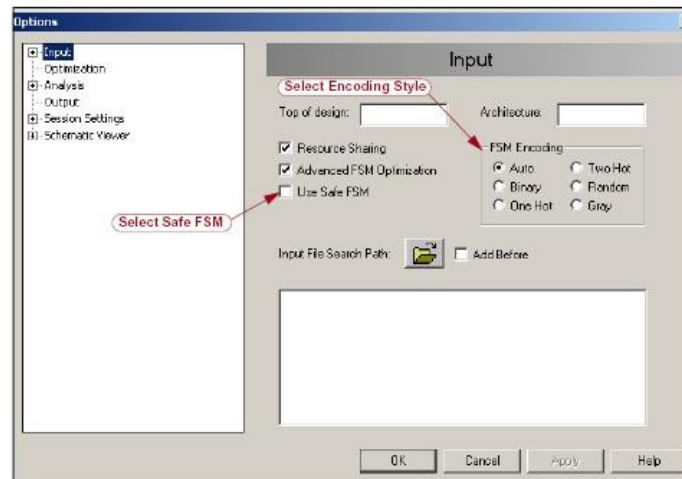
Write a simple Verilog testbench that demonstrate a working flow of the above state machine
Simulate the design

C. Synthesis

Synthesize your Verilog state machine:

- Target technology: Xilinx Artix-7 smallest device and smallest package.
- Timing Constraint: 100MHz

In order to fulfill the requirements of this section you will need to control the state encoding scheme of Precision through the **Tools -> Set Options...Input** dialog box from Precision's GUI (the following picture is taken from Precision's user manual "as is"):



- Run several synthesis iterations. Each one forced to synthesize a different *state encoding scheme*, this way covering all possible *state encoding schemes* in Precision. Identify the state encoding results inside the synthesis log and report (these must be included in your report). **For this particular lab exercise please don't select the Use Safe FSM controls.**

Requirement: The log file and report of all possible *state encoding algorithms* must be included in your report.

D. Place & Route and Gate Level Simulation

Note: In Synthesis and Vivado, target to the Artix7 device of the Basys3 board. Use PMOD pins as I/Os

1. The design that you place & route is as follows:
 - a) Verilog states encoding of "Binary Sequential"
 - b) Precision synthesis of "One Hot"
2. Place & Route the design, using Vivado and generate Verilog netlist and SDF.
3. Perform post place & route gate level simulation in ModelSim – capture the waveform to show that gate level is implemented in "One Hot" encoding

D. Your report must include:

1. Verilog Sources of both design and testbench
2. Source simulation Waveforms
3. Synthesis log files and related place & route screenshots of section D
4. Post place & route simulation waveforms, as required in section D
5. Any other answers to requirements and questions given in the above lab tasks.